

Lecture(1)

Chapter(1)

Logical Thinking

- Introduction
- Formal Logic.
- Connectives and Propositions.

The goal of this chapter is to help you communicate mathematically by understanding the basics of logic

Introduction

Why logic is important in computer science?

- Logic is the Calculus of Computer Science. A computer is a machine that processes data into information using logic.
- So the study of Logic is essential for CS, since Logic is involved in broad range of intellectual activities and it is a base in many areas of computer science such as artificial intelligence, algorithms etc.

Introduction

- Logic is a systematic way of thinking that allows us to deduce new information from old information and to parse the meanings of sentences.
- You use logic informally in everyday life and certainly also in doing mathematics.
- For example, suppose you are working with a certain circle, call it "Circle X," and you have available the following two pieces of information.
 1. Circle X has radius equal to 3.
 2. If any circle has radius r , then its area is πr^2 square units.
You have no trouble putting these two facts together to get:
 3. Circle X has area 9π square units.
- In doing this you are using logic to combine existing information to produce new information. Because deducing new information is central to mathematics, logic plays a fundamental role.

Introduction

Notation

- Is an important part of mathematical language.
- We translate a problem to notation and then perform well-defined symbolic manipulations on that notation.
- This is the essence of the powerful tool called formalism.
- One nice feature of formalism is that it allows you to work without having to think about what all the symbols mean. In this sense, formal logic is really "logical not-thinking".

Example: If we have 5 employees with a monthly salary 1000 SA for each one, then the total amount of money they earned is to 5000.

By formalism it is only the multiplication of 5 by 1000 i.e

T: total amount of money

E: number of employees

S: salary for each one

$$T = ES$$

Connectives and Propositions

- In order to formalize logic, we need a system for translating statements into symbols. We will start with a precise definition of statement.

Definition: A statement (also known as a **proposition**) is a declarative sentence that is either true or false, but not both.

Example:

- 7 is odd.
 - $1+1 = 4$
 - If it is raining, then the ground is wet.
- Note that we don not need to be able to decide whether a statement is true or false in order for it to be a statement.

Connectives and Propositions

- How can a declarative sentence fail to be a statement?
declarative sentence may contain an unspecified term:

x is even.

The truth of the sentence depends on the value of x , so if that value is not specified, we can not regard this sentence as a statement.

Examples:

Statements

- July is the first month of the year. (this is a declarative sentence which is false).
- January is the first month of the year. (this is a declarative sentence which is true)
- The number 2 is even. (this is a declarative sentence which is true).

Non-Statements

- What time is it? (just a question)
- Red is pretty. (we can't decide)
- $2x+10=14$. (x is unknown so we can't decide)

Exercises

Decide whether or not the following are statements. In the case of a statement, say if it is true or false, if possible.

1. $x + 3$
2. $x + 3 = 4$
3. If x and y are real numbers and $5x = 5y$, then $x = y$.
4. The equation $x + 2 = 3$ has exactly one solution.
5. The equation $x + 2 = 3$ has more than one solution.

Connectives and Propositions

Propositions can be combined with connectives such as (and) and (implies to) create compound propositions.

Example:

1. 2 is prime number and $4 + 6 = 10$.
2. Today it is raining implies that tomorrow the sun will shine.

Writing out the entire text of a compound proposition can be tedious, particularly if it contains several propositions. As a shorthand, we will use: lower case letters (like a, b, c, etc.) for simple propositions, and UPPER CASE LETTERS (like A, B, C, etc.) for compound propositions.

Propositions can be true or false. If we know what truth value to assign one we can utilize this information. Otherwise, we check what happens when the proposition is assumed to be true and then false by using a truth table. The following truth tables reveal the meaning of the various connectives.

Connectives and Propositions

Connectives:

- The symbols $\neg$, $\wedge$, $\vee$, $\Rightarrow$ and $\Leftrightarrow$ are called propositional connectives.
- Their properties are best shown via truth tables.
- Note that (T = True. F = False).

Example: If p is the statement "you are wearing shoes" and q is the statement "you can't cut your toenails," then

$$p \rightarrow q$$

- represents the statement, "If you are wearing shoes, then you can't cut your toenails."
- We may choose to express this statement differently in English: "You can't cut your toenails if you are wearing shoes," or "Wearing shoes makes it impossible to cut your toenails."

Connectives and Propositions

Negation: (symbol: $\neg$)

- Interpretation: $\neg a$ means "not a".

a	$\neg a$
T	F
F	T

- Column a has all possible Truth values, however column $\neg a$ indicates the Truth values for not a .

Connectives and Propositions

Conjunction: (symbol: $\wedge$)

- Interpretation: $a \wedge b$ means "a and b".

a	b	$a \wedge b$
T	T	T
T	F	F
F	T	F
F	F	F

- Let a and b be propositions. The proposition "a and b" denoted by $a \wedge b$, is the proposition that is true when both a and b are true and is False otherwise.
- The proposition $a \wedge b$ is called conjunction of a and b .

Connectives and Propositions

Disjunction: (symbol: $\vee$)

- Interpretation: $a \vee b$ means "a or b".

a	b	$a \vee b$
T	T	T
T	F	T
F	T	T
F	F	F

- disjunction is true whenever at least one of the propositions is true. This connective is sometimes called inclusive or to differentiate it from exclusive or (which is often denoted by $+$).
- The formula $a + b$ is interpreted as "or , but not both".

Connectives and Propositions

Implication: (symbol: $\rightarrow$)

- Interpretation: $a \rightarrow b$ means "if a then b" (in the mathematical sense).

a	b	$a \rightarrow b$
T	T	T
T	F	F
F	T	T
F	F	T

- Let a and b be propositions. The implication $a \rightarrow b$, is the proposition that is false when a is true and b is false and true otherwise.
- In this implication a is called the hypothesis and b is called the conclusion.

Connectives and Propositions

Biconditional: (symbol: $\leftrightarrow$)

- Interpretation: $a \leftrightarrow b$ means "a if and only if b"

a	b	$a \leftrightarrow b$
T	T	T
T	F	F
F	T	F
F	F	T

- The biconditional is true exactly when the propositions have the same truth value.
- In some texts, the phrase "is a necessary and sufficient condition for b" is used for $a \leftrightarrow b$.

Exercises

Express each statement as one of the forms $P \wedge Q$, $P \vee Q$, or $\neg P$. Be sure to also state exactly what statements P and Q stand for.

1. The number 8 is both even and a power of 2.
2. $x \neq y$
3. There is a quiz scheduled for Wednesday or Friday.

Without changing their meanings, convert each of the following sentences into a sentence having the form "If P , then Q ."

1. Whenever people agree with me I feel I must be wrong.

Without changing their meanings, convert each of the following sentences into a sentence having the form " P if and only if Q ."

1. If $xy = 0$ then $x = 0$ or $y = 0$, and conversely.